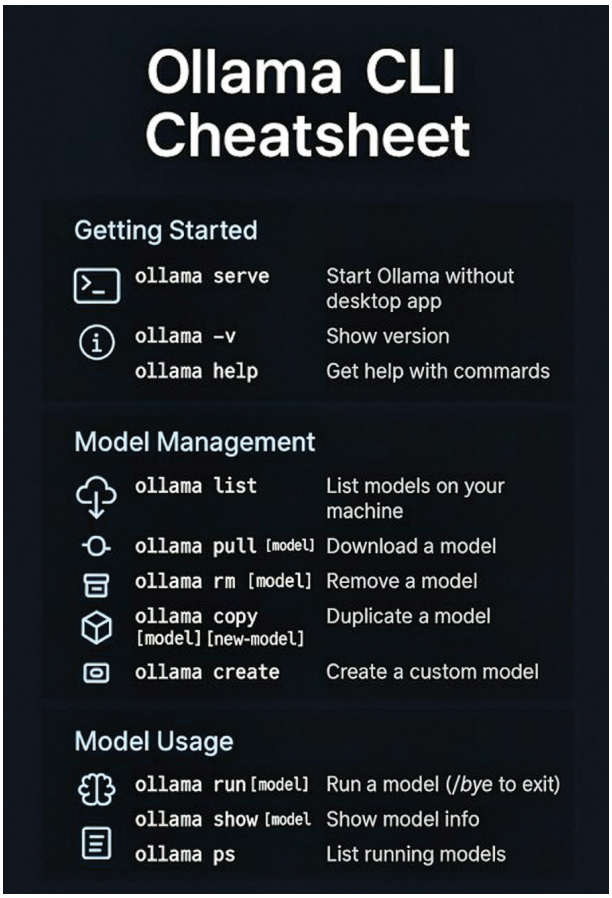


Current Ollama versions open an interactive menu, while direct commands still work normally.

Step 2: Download and run the same GGUF

Hugging Face's Qwen3-4B GGUF page provides a convenient direct Ollama route. Run:

```
ollama run hf.co/ggml-org/Qwen3-4B-GGUF:Q4_K_M
```

The first run downloads the model; subsequent runs use the local copy. If you prefer Ollama's own model library, the same basic pattern is `ollama run model-name`. For example, Ollama currently lists Qwen3.5 4B as a 3.4GB Q4_K_M model, but use our Qwen3-4B GGUF during this chapter if you want a fair cross-runtime comparison.

When the prompt marker appears, paste the same battery question.

Type `/bye` when you want to exit the interactive chat.

Step 3: See what is installed

Run:

```
ollama ls
```

This gives you your local model list. Use it often; local AI users accumulate weights surprisingly quickly.

To remove a model you no longer need:

```
ollama rm <model-name>
```

Do not remove our test model yet.

Step 4: Check whether the GPU is really doing the work

While the model is loaded, open another terminal and run:

```
ollama ps
```

The output shows running models, allocated context and a PROCESSOR field that can reveal whether execution is on the GPU, CPU or split across both.

This is more useful than assuming that "I have an RTX GPU, therefore Ollama must be using it". If you discover a CPU/GPU split, the model may be larger than available VRAM or your current setup may be offloading part of the workload.

Ollama's current context defaults also vary with VRAM: below 24GiB it uses 4K, from 24–48GiB it defaults to 32K, and at 48GiB or above it defaults to 256K. Larger context requires more memory. Those defaults may change in future versions, so treat ollama ps as the truth about your running session.

Step 5: Stop a loaded model

Ollama can keep a model loaded for reuse. To explicitly stop it:

```
ollama stop <model-name>
```

Then run `ollama ps` again. It should disappear from the running list.

This distinction – installed versus loaded – is the same one you saw in LM Studio. A model can occupy gigabytes on disk without consuming its full inference memory.

Step 6: Make a reusable personal preset

Ollama's Modelfile is a simple blueprint for creating a configured variant of a model. This is not fine-tuning; you are defining runtime settings and instructions.

For a straightforward demonstration, pull an Ollama-library model such as:

```
ollama pull qwen3.5:4b
```

Create a text file named `Modelfile` containing:

```
FROM qwen3.5:4b
PARAMETER temperature 0.6
PARAMETER num_ctx 4096
SYSTEM You are a concise technical assistant. State uncertainty clearly and avoid invented facts.
```

Save it, then create your configured variant:

```
ollama create my-local-assistant -f Modelfile
```

Run it:

```
ollama run my-local-assistant
```

You now have a reusable local preset with a system instruction and settings attached. Ollama's Modelfile supports parameters including context length, temperature, top-k, top-p, stop sequences and more. Resist the temptation to tweak all of them at once.